Version 7.4.2

PVsyst - Simulation report

Grid-Connected System

Project: Abhipsa Chakraborty

Variant: Project 4 22 Tilt -21.6 Orientation

No 3D scene defined, no shadings

System power: 30.2 kWp

Entrance Groningen - Netherlands

Author

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Project summary

Geographical Site
Entrance Groningen
Netherlands

Situation

Latitude	53.22 °N
Longitude	6.57 °E
Altitude	11 m
Time zone	UTC+1

Project settings

Albedo	0.20

Meteo data
Eelde
Custom file - Imported

System summary

Grid-Connected System

No 3D scene defined, no shadings

PV Field Orientation
Fixed plane
Tilt/Azimuth 20 / -24 °

Near Shadings
No Shadings

User's needs
Unlimited load (grid)

System information

PV Array

Nb. of modules	114 units
Pnom total	30.2 kWp

Inverters

Nb. of units	1 unit
Pnom total	25.00 kWac
Pnom ratio	1.208

Results summary

Produced Energy	32992 kWh/year	Specific production	1092 kWh/kWp/year	Perf. Ratio PR	86.96 %

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General parameters

Grid-Connected System — **No 3D scene defined, no shadings**

PV Field Orientation

Orientation
Fixed plane
Tilt/Azimuth 20 / -24 °

Sheds configuration
No 3D scene defined

Models used
Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon
Free Horizon

Near Shadings
No Shadings

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	REC
Model	REC 265PE
(Original PVsyst database)	
Unit Nom. Power	265 Wp
Number of PV modules	114 units
Nominal (STC)	30.2 kWp
Optimizer Array	3 Strings x 19 In series
At operating cond. (50°C)	
Pmpp	27.20 kWp
Output of optimizers	
Voper	750 V
I at Poper	36 A

Inverter

Manufacturer	SolarEdge
Model	SE25K-ISR
(Original PVsyst database)	
Unit Nom. Power	25.0 kWac
Number of inverters	1 unit
Total power	25.0 kWac
Operating voltage	750 V
Pnom ratio (DC:AC)	1.08

SolarEdge Power Optimizer

Model	P600 for SE15k+
Unit Nom. Power	600 W
Input modules	2 in series

Total PV power

Nominal (STC)	30 kWp
Total	114 modules
Module area	188 m²
Cell area	166 m²

Total inverter power

Total power	25 kWac
Number of inverters	1 unit
Pnom ratio	1.21

Array losses

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 279 mΩ
Loss Fraction 1.5 % at STC

LID - Light Induced Degradation
Loss Fraction 1.5 %

Module Quality Loss
Loss Fraction -0.5 %

Module mismatch losses
Loss Fraction 0.5 % at MPP

IAM loss factor
Incidence effect (IAM): User defined profile

0°	30°	45°	60°	70°	75°	80°	85°	90°
1.000	1.000	1.000	0.974	0.907	0.832	0.688	0.445	0.000

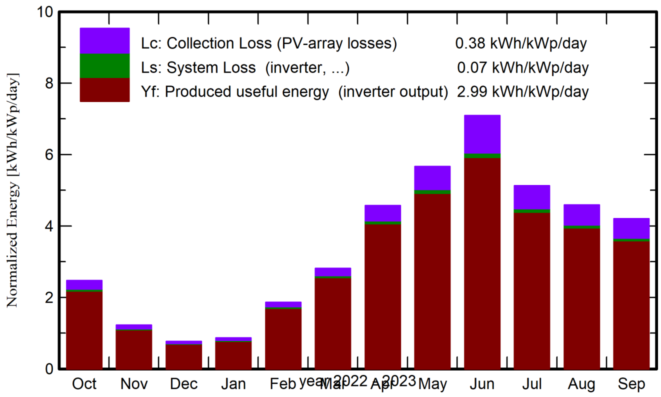
PVsyst V7.4.2
VC8, Simulation date:
17/10/23 13:39
with v7.4.2

Main results

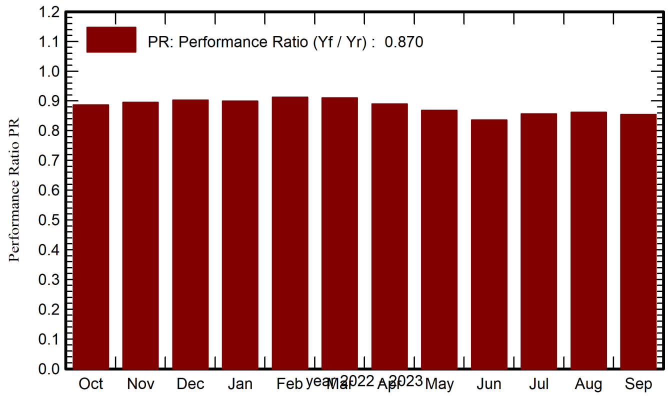
System Production

Produced Energy	32992 kWh/year	Specific production	1092 kWh/kWp/year
		Perf. Ratio PR	86.96 %

Normalized productions (per installed kWp)



Performance Ratio PR



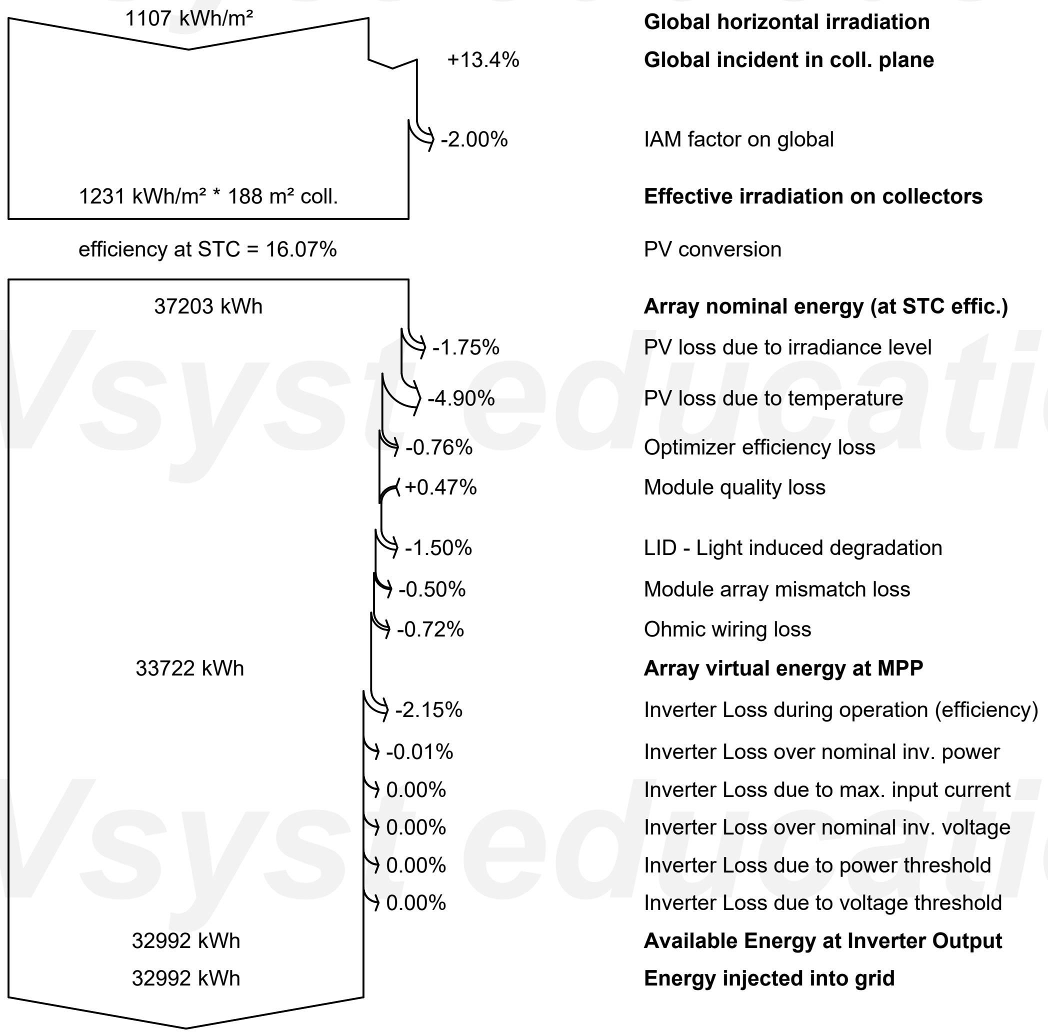
Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m²	kWh/m²	°C	kWh/m²	kWh/m²	kWh	kWh	ratio
Oct. 22	58.6	34.35	12.57	76.5	74.9	2094	2048	0.886
Nov. 22	26.2	17.97	7.86	36.7	35.7	1020	994	0.896
Dec. 22	15.4	10.59	3.17	23.8	23.0	667	648	0.903
Jan. 23	19.0	13.97	5.20	26.8	26.0	750	728	0.900
Feb. 23	38.5	22.73	5.09	52.2	50.9	1473	1438	0.912
Mar. 23	72.6	44.34	6.10	87.3	85.5	2453	2399	0.910
Apr. 23	120.5	57.14	8.07	137.0	134.4	3762	3683	0.890
May 23	165.4	75.62	12.35	175.7	172.6	4709	4612	0.869
June 23	202.6	68.93	18.04	212.7	209.1	5486	5374	0.836
July 23	153.4	86.95	17.46	159.0	155.8	4202	4115	0.856
Aug. 23	130.2	73.15	16.76	142.1	139.3	3779	3700	0.862
Sep. 23	104.9	49.11	16.95	126.1	123.7	3324	3255	0.855
Year	1107.3	554.85	10.83	1255.9	1230.8	33718	32992	0.870

Legends

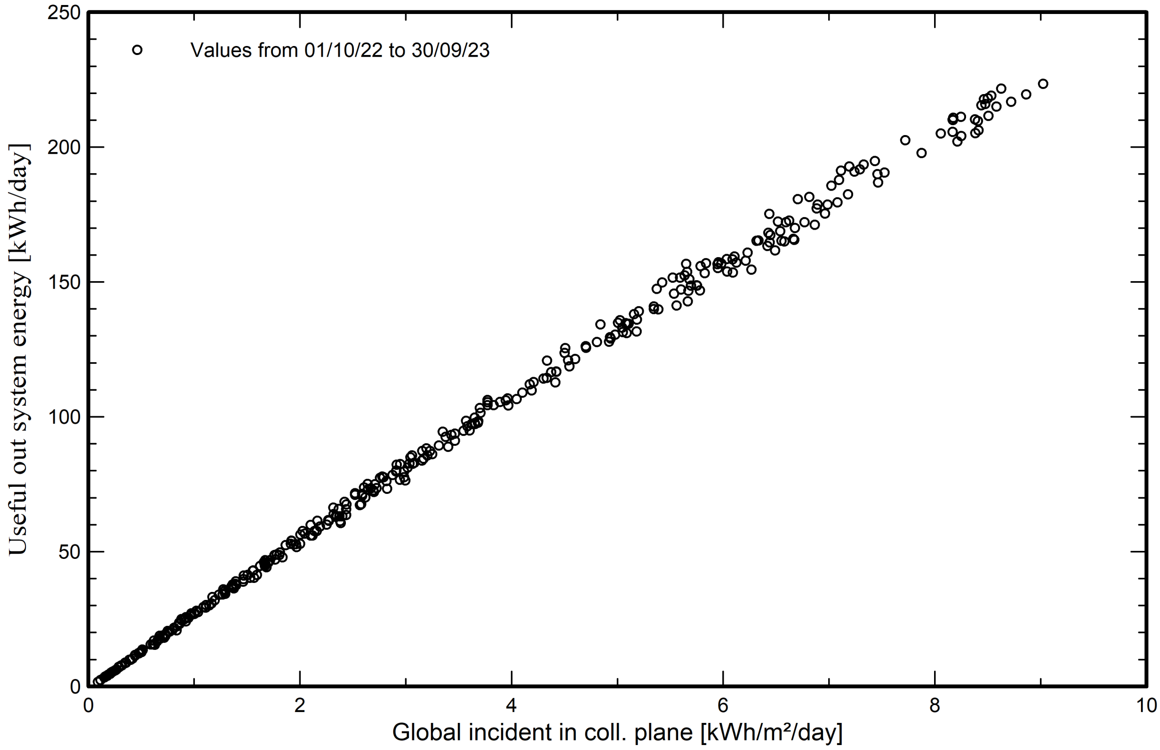
GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane		
GlobEff	Effective Global, corr. for IAM and shadings		

Loss diagram



Predef. graphs

Daily Input/Output diagram



System Output Power Distribution